

TA(s) Consulted: [Name(s) of TA(s) you spoke to]

Instrument Cataloguer

1 Team Members

- Full Name 1 – Student ID – email@mcmaster.ca
- Thanihan Parasutharan – 400583703 – parasutt@mcmaster.ca
- Full Name 3 – Student ID – email@mcmaster.ca

2 Task Overview

[Describe the project at a high level. State the task title, what problem you are solving, why it matters, and what makes it non-trivial/challenging.] We are creating a program to catalogue various percussive instruments based on their waveform. Many musicians find it very difficult to catalogue instruments because many instruments do not have physical identifiers on them.

This task is challenging because: the raw waveform must be converted with a Fourier transformation into time–frequency features such as a spectrogram.

3 Task Definition

[State clearly: What type of data are you using (text, image, tabular, audio, etc.)? Is this a classification, regression, or generation task? If classification, how many classes, and is it single-label or multi-label?] The data we are working with is an audio .wav file. This is a single-label, multi-class classification task with 8 classes, one per instrument in the dataset: flute, oboe, cor anglais (english horn), clarinet, bass clarinet, bassoon, contrabassoon, saxophone, french horn, trombone, trumpet, tuba, banjo, cello, db bass, guitar, mandolin, viola, violin. Each clip is one note by one instrument. The input features describe the sound of the note: fundamental frequency, timbre, harmonics, resonance, dynamics. We have 39 classes (various percussive instruments). We have multiple features: Fundamental frequencies, Timbre, Resonance, Harmonics, noisiness/dynamics

4 Problem Description and Impact

[Explain the problem in more depth, its real-world significance/impact, and elaborate on the technical or practical challenges involved.] Suppose a recording of a composition uses a certain wind/string instrument, and we want to find the same instrument again. Being able to say what instrument a sound came from based on how it sounds makes it easier to search.

The challenges are: (1) audio analysis requires knowledge of signal-processing, we will be using Fourier transforms; (2) the same instrument sounds different across pitches, dynamics and articulation, so the model must not just learn "pitch" or "loudness" as a shortcut; (3) many instruments under the same family share very similar timbre; (4) because the recordings are really clean this may not transfer to real world recordings.

5 Data Source(s) and Collection Plan

[Describe where your data comes from. Include:

- Links to the data source(s):
- <https://philharmonia.co.uk/resources/sound-samples/>.
- Metadata: each file's name has the instrument, pitch, note duration, dynamics and articulation
- Data is already labeled
- If not collected by you: details on how it was originally collected/annotated/preprocessed
- We will be using a subset of the full database (brass, string, woodwinds). This is because percussion instruments do not have that many recordings.

6 Dataset Size and Example Data Points

[State the expected number of data points in your dataset (minimum 1,000).] The collection of instruments has well over 1000 data points (one note per file), once we parse the data we will report the exact count

Example Data Points

1. **Input:** [flute_A4_1_forte_normal.wav (a 1-second flute note A4, forte)] **Label:** [flute]
2. **Input:** [clarinet_C4_1_piano_normal.wav (clarinet note, C4, piano)] **Label:** [clarinet]
3. **Input:** [bassoon_G2_1_fortissimo_normal.wav (low bassoon note, G2)] **Label:** [bassoon]

7 Proposed Solution

[Describe your planned approach:

- each clip is resampled, trimmed, normalized, then converted to a spectrogram. The target label is one of the 19 instruments
- What model(s) will you try (no simple linear regression)?
- instrument recognition is well studied, Our project applies these ideas into a focused instrument subset.

- How will you evaluate whether your model is good (metrics, validation strategy)?
- librosa, torchaudio/PyTorch, scikit-learn, pandas, NumPy, and matplotlib

Sources of Inspiration

1. J. Engel et al., “Neural Audio Synthesis of Musical Notes with WaveNet Autoencoders” (the NSynth dataset), ICML 2017.
2. J. J. Bosch, J. Janer, F. Fuhrmann, and P. Herrera, “A Comparison of Sound Segregation Techniques for Predominant Instrument Recognition in Musical Audio Signals” (the IRMAS dataset), ISMIR 2012.
3. Y. Han, J. Kim, and K. Lee, “Deep Convolutional Neural Networks for Predominant Instrument Recognition in Polyphonic Music,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2017.
4. Q. Kong et al., “PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2020.

8 TA Consultation Summary

[Summarize what you discussed with your TA, the advice/feedback they provided, and how it influenced or changed your project plans.]